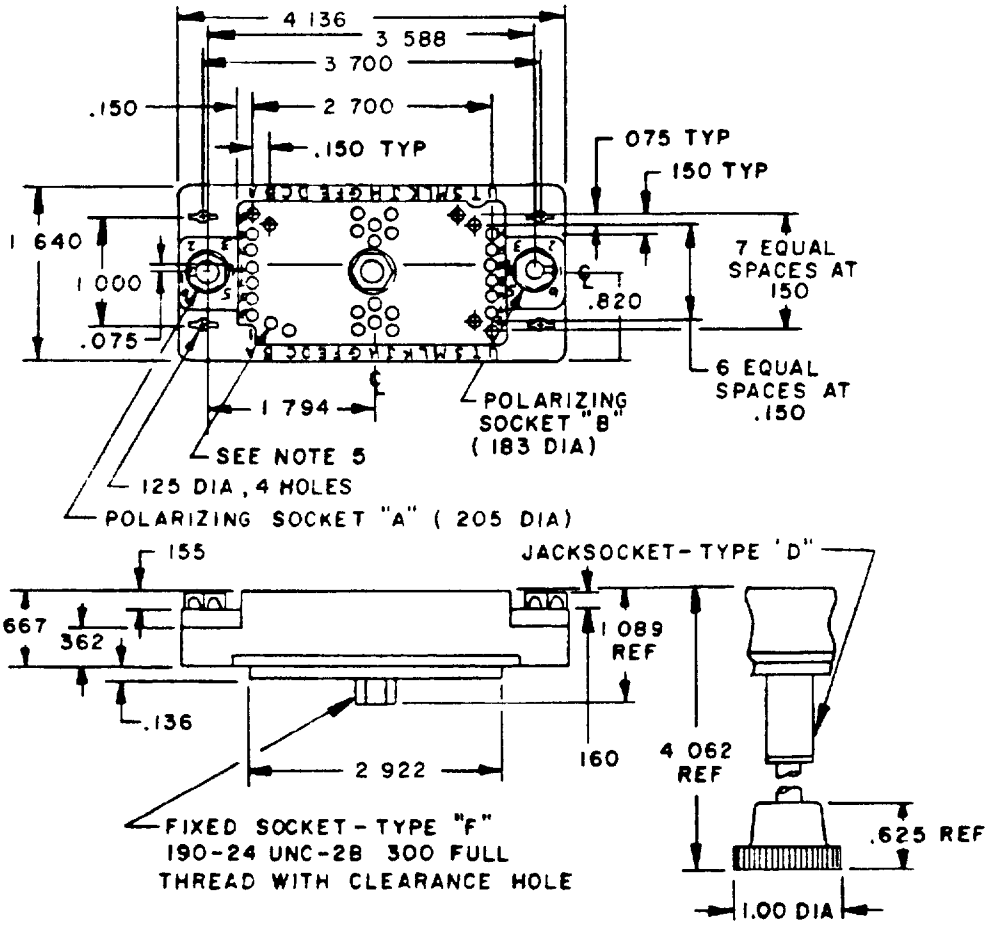
MIL-C-28731/6E
4 JUNE 1985
SUPERSEDING
MIL-C-28731/6D
25 May 1984

MILITARY SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, INSERT (INSULATOR), FEMALE RECTANGULAR, POLARIZED, CENTER SCREW LOCK, FOR 130 REMOVABLE CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

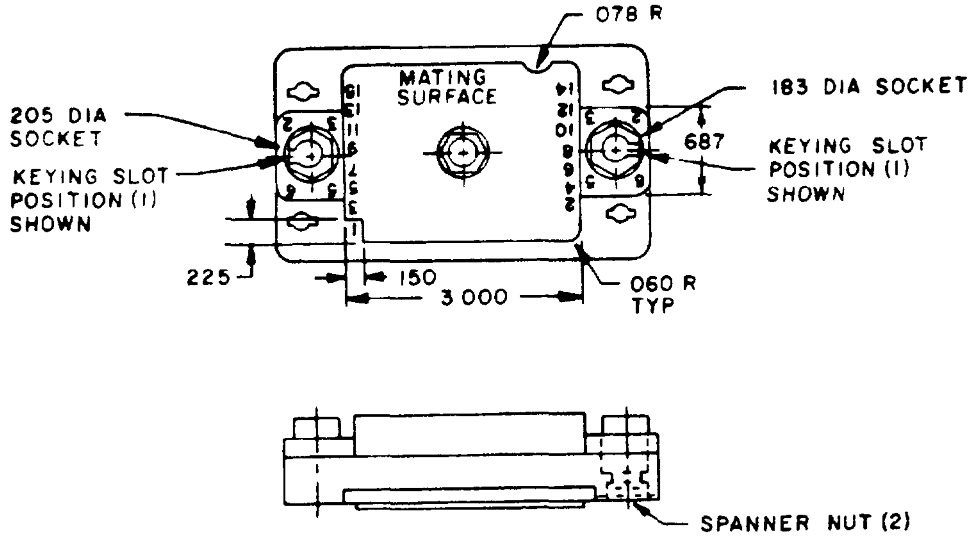
The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-28731.



INCHES	MM
.075	1.91
.125	3.18
.136	3.45
.150	3.81
.155	3.94
.160	4.06
.183	4.65
.190	4.83
.205	5.21
.300	7.62
.362	9.19
.625	15.88
.667	16.94
.820	20.83
1.000	25.40
1.089	27.66
1.640	41.66
1.794	45.57
2.700	68.58
2.922	74.22
3.588	91.14
3.700	93.98
4.062	103.17
4.136	105.05

FIGURE 1 Connector, female rectangular

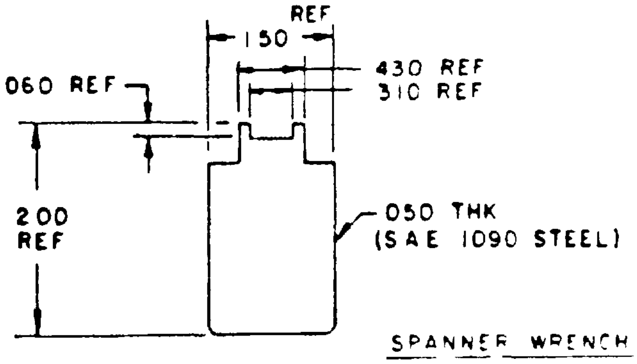
Ⓔ denotes changes



POLARIZING POSITION CHANGE

Keying position on both polarizing sockets can be rotated to desired position by loosening spanner nut (counter-clockwise), resetting socket in one of six positions then retightening spanner nut (clockwise).

NOTE Overtightening of spanner nut can fracture connector. Spanner keying wrench shown below.

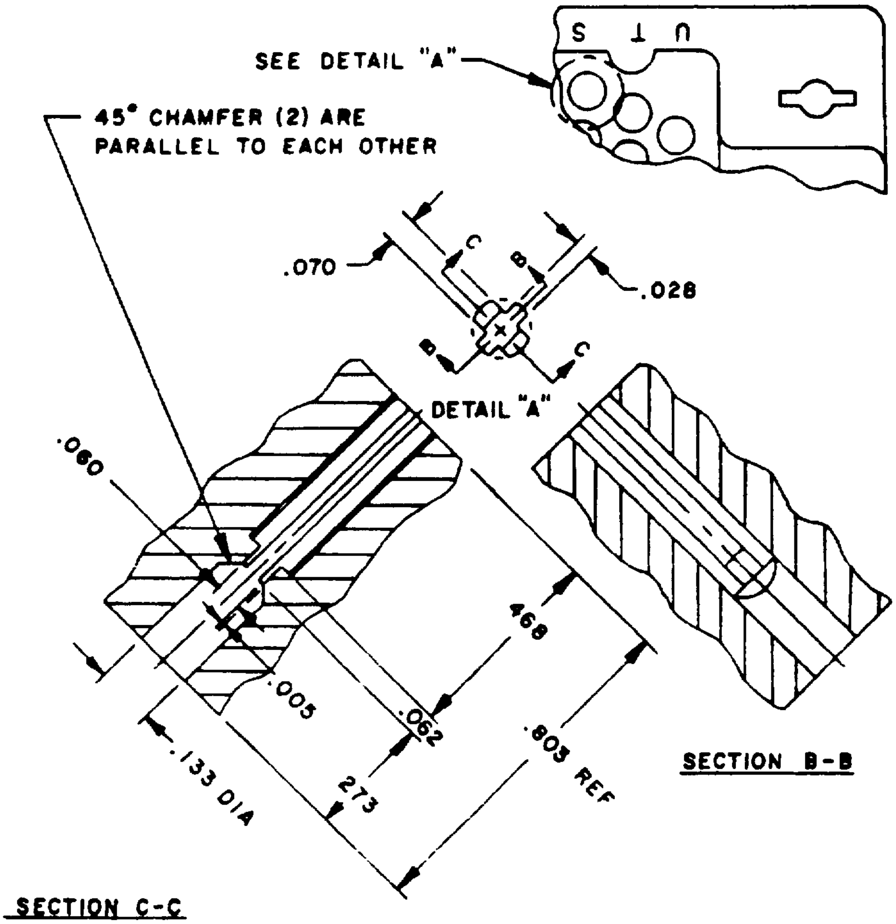


INCHES	MM
.050	1.27
.062	1.57
.078	1.98
.150	3.81
.187	4.75
.225	5.71
.310	7.87
.430	10.92
.687	17.45
1.500	38.10
[illegible]	[illegible]
3.000	76.20

Configuration optional

Configuration - connector polarization guide location

FIGURE 1. Connector, female, rectangular - continued.



Configuration B - Contact hole detail

INCHES	MM
.005	0.13
.028	0.71
.060	1.52
.062	1.57
.070	1.78
.133	3.32
.273	6.93
.468	11.89
.803	20.40

FIGURE 1. Connector, female rectangular - Continued.

NOTES

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerances between any two contact centers shall be ± .004 (0.10 mm).
4. Flatness, squareness, and parallelism shall be within dimensional tolerances.
5. Contact hole details shall be in accordance with configuration B.
6. Mating connector information:
   M28731/6-0073, -0075, and -0077 mate with M28731/5-0109, -0110, -0112 and -0113.
   M28731/6-0074 and -0076 mate with M28731/5-0111, -0114 and 0115.
7. Polarization information:
   Polarization sockets are provided for indexing to preclude intermixing of adjacent connectors. By using a standard type spanner wrench, polarization sockets can be rotated to desired position. For parts supplied directly to the government, polarization will be in the 1-1 position. For other polarization positions see connector polarization guide configuration A (see marking, note 13d). For direct government acquisition a tool is supplied in the unit package for polarization socket adjustment.
8. Contact and terminal designations:
   Contacts and terminals shall be designated on the insulator using letters in the location shown on figure 1.
9. Tolerance shall be ±.010 (0.25 mm) unless otherwise specified.
10. Patent notice.
    Patent No. 3,248,686 owned by Elco Corporation. The government has a royalty-free license under the above patent for the benefit of manufacturers of items called for in this specification, either for the government or for use in equipment to be delivered to the government.
11. For direct government acquisition crimp contacts M28731/20-0001 are furnished in a plastic bag in each unit package (see marking, note 13a).
12. For direct government acquisition solder hole and wrappost contacts M28731/25-0001 and /26-0001 are factory installed in connector body (see marking, note 13a).
13. Marking:
    - a. Connectors supplied directly to the government shall be marked with the military part number (see table I). Example: M28731/6-0074.
    - b. Connectors supplied with contacts to government and users shall be marked with the military part number (see table I). Example: M28731/6-0074.
    - c. Connectors supplied without contacts to government and users shall be marked as follows. Example: M28731/6D (screw or socket type from table I). The user shall complete the remaining portion of the military part number as shown in table I based upon contact type he assembles. Example: M28731/6D 0074.
    - d. Polarization position marking is not required unless otherwise specified. When required polarization marking shall be in accordance with the connector polarization guide (see configuration A). The marking shall appear following the part number.
14. Applicable contact insertion and removal tools:

    Insertion MIL-I-81969/13-[illegible]
    Removal MIL-I-81969/11-01

15. When double wire crimp contacts M[illegible] is required for use with the connector, it shall be ordered separately.
16. Rectangular [illegible] is [illegible] for [illegible] MIL-C-28731/[illegible] or -0006 and must be ordered separately.

(E)

FIGURE 1. Connector, female rectangular - Continued.

REQUIREMENTS.

Dimensions and configuration See figure 1.

Mating and unmating forces (torque) 150 inch-ounces, maximum.

Mechanical shock (specified pulse) Applicable.

Part number See table I.

TABLE I. Part number information.

Military part number	Actuating hardware	Polarizing socket position		Contact type
		A .235 Dia.	B .183 Dia.	
M28731/6-0073	Fixed socket Type F	1	1	Crimp MIL-C-28731/20 1/
M28731/6-0074	Jack socket Type D	1	1	Crimp MIL-C-28731/20 1/
M28731/6-0075	Fixed socket Type F	1	1	Solder hole MIL-C-28731/25 2/
M28731/6-0076	Jack socket Type D	1	1	Solder hole MIL-C-28731/25 2/
M28731/6-0077	Fixed socket Type F	1	1	Wrappost MIL-C-28731/26 2/

1/ See figure 1, note 11
2/ See figure 1, note 12

NOTE
M28731/6-0073 supersedes M28731/6-0001 thru M28731/6-0036
M28731/6-0074 supersedes M28731/6-0037 thru M28731/6-0072

Custodians
Army - CR
Navy - EC
Air Force - 85

Review activities
Army - AR, AT, MI
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AL, AV, ME
Navy - MC, OS
Air Force - 19

Agent
DLA - ES

Preparing activity
Navy - EC

(Project 5935-3498-5)